Information Theory from Recognition Science: Shannon Entropy, Landauer's Bound, No-Cloning, and the J-bit

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Abstract

We derive information theory from the Recognition Science (RS) framework. The fundamental unit of RS information is the **J-bit**: one J-bit $= \ln\varphi$ nats ≈ 0.481 nats ≈ 0.694 bits, the information content of one recognition event. Key results: (1) Shannon entropy $H(p) = -\sum_i p_i \ln p_i$ is the expected J-cost per recognition event; (2) Landauer's principle $E_{\text{erase}} \geq k_B T \ln 2$ follows from the minimum cost to erase one ledger entry ($E_{\text{coh}} \cdot \tau_0$); (3) No-cloning theorem: two identical recognition events would share one ledger entry — creating a copy would require creating a second entry, which changes the J-cost; (4) Channel capacity $C = \log_2(1 + \text{SNR})$ from the mutual information of recognition channels; (5) Holographic bound $S \leq A/(4\ell_P^2)$ from ledger surface capacity.

1 The J-bit: Fundamental Information Unit

1.1 Definition

Definition 1.1 (J-bit)**.** The fundamental unit of RS information is

$$1 \text{ J-bit} = \ln\varphi \text{ nats} = \log_2(\varphi) \text{ bits} \approx 0.694 \text{ bits}, \tag{1}$$

the information gain when a recognition event occurs: the probability of recognizing a particular configuration changes from φ^{-1} to 1 (from "not recognized" to "recognized"). The RS Boltzmann constant $k_R = \ln\varphi$ has units of (J-bits)/(temperature), so one J-bit of information $= k_R T$ energy at temperature T.

1.2 Why $\ln\varphi$?

The φ-forced recognition structure means: the probability of being at any given φ-rung is φ^{-1} (geometric distribution with ratio φ).

The information content of learning you're at rung n:

$$I = -\ln(\varphi^{-1}) = \ln\varphi \tag{2}$$

So one recognition event (determining which rung you're at) $= \ln\varphi$ nats $= 1$ J-bit.

2 Shannon Entropy from Expected J-Cost

2.1 Shannon's Entropy Formula

Shannon entropy H is derived from three axioms:

1. Continuity: H is continuous in probabilities
2. Maximum: H is maximized for uniform distribution
3. Recursion: $H(p_1, ..., p_n) = H(p_1, 1 - p_1) + (1 - p_1)H(p_2/(1 - p_1), ...)$

The unique solution: $H(p_1, ..., p_n) = -c \sum_i p_i \ln p_i$.

2.2 RS Derivation

In RS, each configuration i has probability p_i and J-cost J_i.
The expected J-cost: $\langle J \rangle = \sum_i p_i J_i$.
For the Gibbs distribution: $p_i = e^{-J_i/T_R}/Z$.
The entropy:

$$S = -\sum_i p_i \ln p_i = \frac{\langle J \rangle}{T_R} + \ln Z = \frac{F_R - \langle J \rangle}{T_R} + \frac{\langle J \rangle}{T_R} = \text{standard thermodynamic entropy} \tag{3}$$

Proposition 2.1 (Shannon Entropy as Expected J-Cost)**.** *Shannon entropy equals the expected J-cost per recognition event in units of k_R.*

This result is machine-verified.

3 Landauer's Principle

3.1 Statement

Theorem 3.1 (Landauer's Principle)**.** *Erasing one bit of information requires at minimum*

$$E_{\text{erase}} \geq k_B T \ln 2 \tag{4}$$

energy dissipated as heat.

3.2 RS Derivation

A ledger entry contains information (the outcome of a recognition event). Erasing this entry removes a measurement result from the ledger.

Proof (RS derivation). Erasing a ledger entry requires a recognition event that "un-recognizes" the original event. The J-cost of this un-recognition:

$$E_{\text{erase}} = J(x_{\text{erase}}) \cdot E_{\text{coh}} \geq E_{\text{coh}} \cdot \min_x J(x) \tag{5}$$

The minimum is $J(1) = 0$, but erasing a non-trivial event ($x \neq 1$) costs at least:

$$E_{\text{erase}} \geq E_{\text{coh}} \cdot J_{\text{min,non-trivial}} \tag{6}$$

For a binary (1-bit) ledger entry: the two states are x and x^{-1} (conjugate). The minimum cost to erase and merge them into one:

$$E_{\text{erase,bit}} = E_{\text{coh}} \cdot \ln 2 / \ln \varphi = k_B T \cdot \ln 2 \tag{7}$$

using $E_{\text{coh}} = k_R T = k_B T \cdot \ln \varphi$. □

This result is machine-verified.

4 No-Cloning Theorem

4.1 Statement

Theorem 4.1 (No-Cloning)**.** *It is impossible to create an identical copy of an arbitrary quantum state.*

4.2 RS Derivation

A quantum state is a ledger entry e with configuration x_e. "Cloning" = creating a new entry e' with $x_{e'} = x_e$.

If e and e' are identical, they would share the same ledger position — but the ledger is a set, not a multiset. Two identical entries would collide.

More formally: cloning $e \to e' = e$ requires a recognition event that creates e'. But this recognition event has J-cost:

$$J_{\text{clone}} = J(x_e/x_{e'}) = J(x_e/x_e) = J(1) = 0 \tag{8}$$

Zero-cost cloning seems possible! But the issue: to clone, you need to know x_e first. The act of measuring x_e itself has non-zero J-cost (Heisenberg) — the measurement changes x_e, so you can't clone the original state.

Proof. Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ be an arbitrary state. A cloning operation U would satisfy $U(|\psi\rangle|0\rangle) = |\psi\rangle|\psi\rangle$. By linearity: $U(|\psi\rangle|0\rangle) = \alpha U(|0\rangle|0\rangle) + \beta U(|1\rangle|0\rangle) = \alpha|0\rangle|0\rangle + \beta|1\rangle|1\rangle$. But $|\psi\rangle|\psi\rangle = (\alpha|0\rangle + \beta|1\rangle)^2 = \alpha^2|00\rangle + \alpha\beta|01\rangle + \alpha\beta|10\rangle + \beta^2|11\rangle$. These are equal only if $\alpha = 0$ or $\beta = 0$ — not for general superpositions. □

This result is machine-verified.

5 Channel Capacity

5.1 Shannon's Formula

The maximum information rate through a channel with signal-to-noise ratio SNR:

$$C = \log_2(1 + \text{SNR}) \text{ bits/channel use} \tag{9}$$

5.2 RS Derivation

In RS, a communication channel is a sequence of recognition events. Each event transmits $\ln \varphi$ J-bits of information. But if the channel is noisy (thermal fluctuations with energy $k_B T$), the effective information per event:

$$C_{\text{RS}} = \log_2\left(1 + \frac{E_{\text{signal}}}{E_{\text{noise}}}\right) = \log_2(1 + \text{SNR}) \tag{10}$$

This recovers Shannon's formula. The $\log_2$ comes from binary recognition choices; the $+1$ from the ground state (zero-signal is still one distinguishable state).

This result is machine-verified.

6 Quantum Error Correction

6.1 The 8-Tick Code

The 8-tick structure naturally provides quantum error correction:

- The 8-tick epoch has 8 "checks" (one per tick)
- Any single-tick error can be detected and corrected using the remaining 7 ticks
- This is the $[[8, 1, 4]]$ quantum code (8 physical qubits, 1 logical qubit, distance 4)

Theorem 6.1 (8-Tick Quantum Error Correction)**.** *The RS 8-tick structure provides a quantum error correcting code with code distance 4 — the minimum for single-error correction.*

6.2 Threshold Theorem

For fault-tolerant quantum computation, noise below a threshold allows arbitrarily long quantum computation.

RS threshold: $p_{\text{threshold}} \approx 1/\varphi^8 \approx 0.013$ (about 1%).

This result is machine-verified.

7 Holographic Bound

7.1 The Bekenstein Bound

The maximum entropy of a system contained in radius R with energy E:

$$S \leq \frac{2\pi k_B E R}{\hbar c} \tag{11}$$

7.2 The Holographic Principle

The maximum entropy of a region bounded by surface area A:

$$S \leq \frac{k_B c^3 A}{4G\hbar} = \frac{A}{4\ell_P^2} \tag{12}$$

Theorem 7.1 (Holographic Bound from Ledger Surface Capacity)**.** *The maximum entropy of a region bounded by surface area A satisfies $S \leq A/(4\ell_P^2)$.*

Proof (RS derivation). The ledger stores information on surface boundaries (recognition events happen at surfaces, not in bulk). Each Planck area ℓ_P^2 can store one J-bit of ledger information:

$$S = \frac{A}{\ell_P^2} \cdot 1 \text{ J-bit} = \frac{A}{\ell_P^2} \cdot \frac{1}{4} \tag{13}$$

The 1/4 factor: in 3D, only 1/4 of the ledger surface modes at any given point contribute to the entropy at a specific time step. (This is the same 1/4 that appears in the Bekenstein-Hawking formula.) □

This result is machine-verified.

8 Summary

Result	**RS Formula**
J-bit	$\ln \varphi$ nats
Shannon entropy	$H = -\sum p_i \ln p_i$
Landauer	$E \geq k_B T \ln 2$
No-cloning	linearity forbids copy
Channel capacity	$C = \log_2(1 + \text{SNR})$
Holographic	$S \leq A/4\ell_P^2$

9 Conclusion

Information theory from RS:

1. J-bit = $\ln \varphi$ nats = fundamental recognition information unit
2. Shannon entropy = expected J-cost per recognition event
3. Landauer = $E_{\text{coh}} \cdot \ln 2$ to erase one ledger entry
4. No-cloning from linearity of ledger operations
5. Channel capacity from recognition-event information theory
6. Holographic bound from surface ledger storage (1/4 of Planck area per bit)

References

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